

# Daniel Lofeodo

Montreal, Canada

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## Experience

**2025-04 – 2026-01** **Machine Learning Engineer** (Fixed-Term Contract)

*Hydro-Québec, Montreal, Canada*

RL and forecasting agents for hydro-dam releases under extreme, out-of-distribution (1-in-10,000-year) flood conditions.

- Built an end-to-end forecasting pipeline (XGBoost → transformer/Chronos); found supervised baselines failed to generalize out-of-distribution.
- Developed a constrained RL framework (PPO, dynamic reward shaping) that kept water levels within safety bounds in 9 of 9 extreme scenarios, vs. 2 of 9 for SL baselines.
- Outperformed the legacy deterministic tool's max-outflow assumption: 83% fewer unnecessary releases at equivalent or better safety margins.

**2021-06 – 2024-08** **Software Engineer**

*Creaform, R&D, Montreal, Canada*

Ultra precision 3D scanners and software for engineering.

- Developed core features for aerospace-grade 3D inspection software, including geometric tolerancing (circularity, cylindricity, flatness) for validating scanned parts against engineering specifications.
- Built a new industrial pipe inspection product by repurposing internal components, contributing to a high-revenue product line.
- Led UI/UX development for Creaform.OS, improving performance through profiling and delivering customizable interfaces.
- Contributed to large-scale C++ codebases in an Agile environment with CI/CD, code reviews, and continuous refactoring.

## Education

**2024-08 – 2026-05** **M.Sc. Computer Science (Machine Learning)**

*Montreal Institute for Learning Algorithms, University of Montreal, Montreal, Canada*

- 4.1/4.3 GPA

**2017-08 – 2021-05** **B.Eng. Mechanical Engineering**

*Concordia University, Montreal, Canada*

- Dean's List (2021)

## Spotlight Projects

**2026-04 – 2026-05** **Perspective Blind – 3D Spatial Reasoning Benchmark for VLMs**

Benchmark testing VLM spatial reasoning from an imagined, unseen viewpoint using 3D scene reconstructions.

- Built a benchmark testing viewpoint-invariant spatial reasoning across 5 VLM families on ScanNet 3D reconstructions.
- Found lateral-relation accuracy collapses from 83.6% to 8.1% – below random chance – as viewpoint misalignment increases.
- Built eval infra: schema-enforced outputs, consistency/confusion-matrix metrics, cost/latency tracking across providers.

**2026-05 – 2026-07** **Latent SpaceMail – Autonomous Multi-Agent Newsletter Pipeline**

Agentic pipeline delivering a personalized weekly AI news and research briefing; live 2 months.

- Orchestrated 6 Claude agents on GCP (Cloud Run, Pub/Sub, Firestore) in a fan-out/fan-in pipeline across parallel news/arXiv chains.
- Engineered tool-calling, concurrency control, and retry/backoff to score ~800 articles + 250 papers weekly, sustaining a 70%+ open rate.
- Built a Google-auth subscriber site with bilingual, topic-tiered preferences.

## Technical Skills

**Programming Languages:**

Python, C++, C#, SQL, Java, JavaScript, TypeScript

**AI Frameworks:**

PyTorch, AutoML, scikit-learn, xarray, XGBoost, Pandas, Numpy, stable-baselines3, gymnasium

**Visualization Frameworks:**

matplotlib, Seaborn, Plotly

**Web/App Frameworks:**

Streamlit, Flask, FastAPI, React

**MLOps:**

Azure, GCP, AWS, GitHub, Databricks, Docker, Weights & Biases, Tensorboard

**Environments:**

Windows, WSL, Linux

**Project management:**

Asana, Jira, Teams

## Certifications

- AWS MLA (Cloud-native ML)
- Claude Code in Action
- Complete Modern C++ (C++11/14/17)

## Conferences

- [Autonomous Hydroelectric Dam Operation under Extreme Flooding](#), Mila Annual Conference

## Languages

- English (Native)
- French (Native)
- Italian (Advanced - C1)